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Environmental and social drivers of urban garden biodiversity: a PRISMA-ScR review

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Introduction: Urban gardens are increasingly recognized as multifunctional green spaces that support biodiversity and contribute to sustainable urban living. This scoping review explores how environmental and social factors interact to shape biodiversity outcomes in urban gardens across different taxa.

Methods: Following PRISMA-ScR guidelines, we screened an initial pool of 11,900 records retrieved from Web of Science, Scopus, and PubMed. We synthesized findings from 30 peer-reviewed studies published since 2000, including 26 core studies identified through the PRISMA-ScR process and four additional studies from a supplementary update search. For each study, we extracted taxonomic focus, garden types, spatial scales, and environmental and social variables tested as biodiversity drivers, and synthesized patterns across taxa.

Results: Most studies were conducted in Europe and North America and focused on plants and invertebrates in private and community gardens. Environmental variables—such as garden size, plant species richness, vegetation structure, and landscape connectivity—were consistently associated with higher species richness and abundance. Social and cultural factors, including gardeners' motivations, management intensity, cultural background, and socio-economic context, shaped garden management and species assemblages.

Discussion: This review shows that urban garden biodiversity is jointly shaped by environmental heterogeneity and the decisions of diverse gardeners, but evidence remains geographically biased and taxonomically narrow. Future research should adopt more integrative socio-ecological designs, include underrepresented regions and taxa, and use more comparable biodiversity metrics. Policy and planning should recognize gardens as key components of urban green infrastructure and support management practices that enhance both ecological and social benefits.

KEYWORDS

ecosystem services, environmental factors, garden biodiversity, landscape structure, scoping review, socio-cultural factors, urban gardens, urban sustainability

1 Introduction

Due to industrialization and urbanization, the global urbanization rate (the proportion of the population living in urban areas) reached 55.3% in 2018, meaning that over half of the world's population now resides in cities (United Nations, Department of Economic and Social Affairs, Population Division, 2018). Within urban environments, various habitats are becoming fragmented, reducing species' living spaces, and human activities such as transportation and construction have negative impacts on biodiversity (Beninde et al., 2015). Biodiversity, ecosystems, and the essential services they provide are the pillars of life on Earth, including human life (World Health Organization and Convention on Biological Diversity, 2015). Improving urban environments for biodiversity not only benefits the local residents and communities but also aids biological conservation (Davies et al., 2009).

Maintaining functional and diverse urban green spaces is a fundamental element in creating livable cities (World Health Organization and Convention on Biological Diversity, 2015). In many countries, gardens are a key component of urban green spaces (Kabisch and Haase, 2013; Tresch et al., 2019a), playing an important role in preserving natural ecosystems and supporting diverse species (Ibarra et al., 2021). A study by Di Pietro et al. (2018) found that the loss of community gardens due to urbanization would result in a significant loss of ecological continuity. This finding underscores the notion that urban gardens, in their capacity as green spaces, function as pivotal components within ecological networks. In this review, we use the term urban gardens as an umbrella concept that encompasses private residential gardens, home gardens and plots, community gardens, allotment gardens, and small-scale agroforestry gardens located within urban and peri-urban areas. These spaces may be privately or publicly owned, but all share the characteristics of being intensively managed by gardeners and embedded within the urban fabric.

For instance, gardens, as open habitats, were found to support a higher diversity of dung beetle species compared to closed forests (Asha et al., 2021). Specific features located in gardens (nest boxes, water tanks, feeders, and trees) have been observed to increase the diversity of wildlife, such as birds (Davies et al., 2009; Deguines et al., 2020). Home gardens and plots provide enormous potential habitat for pollinators, and small-scale floristic improvements can attract more beneficial insects in fragmented urban landscapes, supporting urban biodiversity, pollination services, and biological control (Griffiths-Lee et al., 2022). Urban community gardens may include many species that are rarely found across the urban landscape, making them important places for preserving and supporting rare species in cities (Ong et al., 2022). Furthermore, Egerer et al. (2024) underscored the significance of urban community gardens as pivotal systems where human and natural needs converge, signifying their potential to offer emotional, social, and educational benefits to community members while fostering biodiversity conservation. In light of urban sprawl, the subsequent increase in urban density, and efforts to prevent the loss of garden spaces, policies are needed to maximize the various individual,

ecological, and social benefits provided by gardens (Young et al., 2020).

The purpose of this study is to investigate how urban gardens positively influence urban biota. To our knowledge, this is the first scoping review that systematically synthesizes quantitative biodiversity outcomes across multiple taxa in urban gardens while jointly considering both environmental and social drivers. Previous reviews have typically focused either on ecosystem services or on specific taxa, whereas our study explicitly maps how physical garden attributes and gardeners' socioeconomic characteristics interact to shape biodiversity. By updating the evidence base to 2025 and integrating recent empirical studies, this review provides an up-to-date framework for biodiversity-oriented garden planning and management in cities. This study poses the following questions: (1) What has been reported in the literature regarding the effects of garden and surrounding environmental conditions on biota? (2) What has been reported in the literature regarding the effects of the social conditions of people associated with the gardens on biota? The scope of the study includes the surveyed biota, the gardens examined, the research methods, and statistically significant results, which will be described. Given the diversity of taxa, garden types, and methodological approaches in this field, a scoping review is particularly appropriate for providing an integrative overview. Our aim is not to test a single *a priori* hypothesis or to calculate pooled effect sizes, but rather to map and synthesize the range of reported relationships between environmental and social drivers and biodiversity outcomes in urban gardens.

2 Methodology

We conducted a scoping review following the PRISMA-ScR guidelines (Tricco et al., 2018) for evidence mapping, not for hypothesis-testing meta-analyses. This design is suitable given the heterogeneity in study aims, biodiversity metrics, and garden types. Our methodology consisted of three main steps. First, we defined search criteria and conducted a systematic database search following PRISMA-ScR guidelines. Second, we applied predefined inclusion and exclusion criteria to select studies that reported quantitative indicators of biodiversity in urban gardens. Third, we performed a supplementary update search for studies published between 2023 and 2025 and integrated the eligible articles narratively as a recent evidence update.

2.1 Search criteria

This document investigates the positive impacts of urban gardens on urban biota, exploring existing literature to answer key questions regarding environmental and social conditions. The study aims to provide a comprehensive overview of how urban gardens contribute to biodiversity and the well-being of urban ecosystems. It will delve into the types of biota affected, the characteristics of the gardens studied, the research methodologies employed, and the statistically significant results obtained. In this

research, a systematic review of the literature available in the Web of Science (WOS), PubMed and Scopus databases were carried out.

We intentionally used broad garden-related search terms —‘(garden) OR (community AND garden) OR (urban AND garden)’—to capture a wide range of publications that potentially addressed biodiversity in urban gardens. Limiting the query to titles, abstracts, and keywords and subsequently applying biodiversity-related inclusion criteria at the full-text screening stage allowed us to identify studies that explicitly reported quantitative biodiversity outcomes, even when ‘biodiversity’ was not used as a search term. However, we acknowledge that not including explicit biodiversity terms (e.g., ‘biodiversity’, ‘species richness’, ‘species diversity’) and additional garden types (e.g., ‘allotment garden’, ‘private garden’) in the initial query may have resulted in the omission of some relevant studies that did not use garden-related terminology.

The search was conducted in June 2023, with the search period from 2000 to 2022. A total of 11,900 articles were retrieved. The methodological process was carried out according to the recommendations in the ‘Preferred Reporting Items for Systematic reviews and Meta-Analyses extension for Scoping Reviews (PRISMA-ScR) Checklist’ (Tricco et al., 2018) and it is attached as a [Supplementary Table S1](#). We did not register the protocol for this review.

2.2 Selection process

The exclusion criteria are duplicate articles, articles written in languages other than English, articles that did not provide full text, articles that did not cover the selected topic. Two reviewers independently screened titles and abstracts, followed by full-text screening against predefined inclusion criteria. Disagreements were resolved through discussion.

After reviewing titles, abstracts, and keywords, we excluded studies on human pathology, genetics, physiology, and marine topics; research on animals, insects, or microorganisms unrelated to gardens; and studies on compost, soil and plant characteristics. From the 1,354 papers that passed the screening process, 88 papers containing the keywords ‘diversity’, ‘biodiversity’, and ‘species’ in their titles, keywords or abstracts were selected for eligibility assessment.

In the full-text eligibility assessment, reviewed by two independent researchers to ensure that papers whose topics were not experimental in nature with respect to species were excluded, including those on food security, food diversity, ion analysis in plants, and analyses of microorganisms or bacteria found in organisms. This review focused on studies that provided quantitative indicators of biodiversity (e.g., species richness, abundance, diversity indices) of wild organisms such as birds, insects, and soil fauna in urban gardens. Studies dealing primarily with food diversity, food security, or human nutrition were excluded, as their objectives centered on agro-economic or dietary outcomes rather than ecological biodiversity. In the end, we selected 26 research papers that address not only the biodiversity of gardens but also the impact of garden components or garden stakeholders ([Figure 1](#)). For each eligible study, we extracted:

- bibliographic information (year, country, region),
- garden type (private, community, allotment, agroforestry, etc.),
- focal taxa (plants, birds, bees, beetles, etc.),
- biodiversity metrics (species richness, abundance, functional diversity, etc.),
- environmental drivers (e.g., garden size, vegetation structure, surrounding land cover),
- social/management drivers (e.g., gardener socio-demographics, management intensity), and
- effect direction (positive, negative, mixed, or non-significant) for each driver–taxon combination.

One reviewer performed the initial extraction, and a second reviewer checked all entries for accuracy and consistency. These data were attached as a [Supplementary Table S2](#). All variables were coded into standardized categories ([Tables 1, 2](#)).

2.3 Recently screened studies as a supplementary update

In addition to our original systematic search covering 2000–2022, we performed a supplementary update search in May 2025 to capture studies published in 2023–2025. This update yielded several candidate studies examining biodiversity drivers in urban gardens and related small-scale green spaces. After screening abstracts and accessible full texts, we formally included the following additional studies in the synthesis due to their alignment with our inclusion criteria: [de Montaigne and Goulson \(2023\)](#); [Liere and Cowal \(2024\)](#); [Márton et al. \(2025\)](#), and [Morpurgo et al. \(2024\)](#). Given the limited number of new papers, we did not revise the original PRISMA flowchart ([Figure 1](#)) but summarized them narratively. And the supplementary studies and their integration are documented in a [Supplementary Table S2](#).

Note: This pragmatic update approach ensures that recent relevant findings are considered without fully re-running the entire screening pipeline under tight time constraints.

Consistent with the PRISMA-ScR guidance for scoping reviews, our aim was to map and synthesize the range of environmental and social drivers of biodiversity in urban gardens rather than to test *a priori* hypotheses or to calculate pooled effect sizes. We therefore did not perform a formal risk-of-bias assessment or meta-analysis. The included studies varied widely in taxa, response variables (e.g., species richness, abundance, functional diversity), sampling designs, and statistical models, which precluded the calculation of directly comparable effect sizes. Instead, we used descriptive statistics and narrative synthesis to summarize the direction and significance of reported relationships between garden attributes and biodiversity outcomes across taxa. For each article, we identified the primary statistical models employed, the area where the study was conducted, the type of garden, and the studied species. Then, we grouped them into the categories presented in [Table 1](#). Frequencies and percentages were calculated in Microsoft Excel (Microsoft

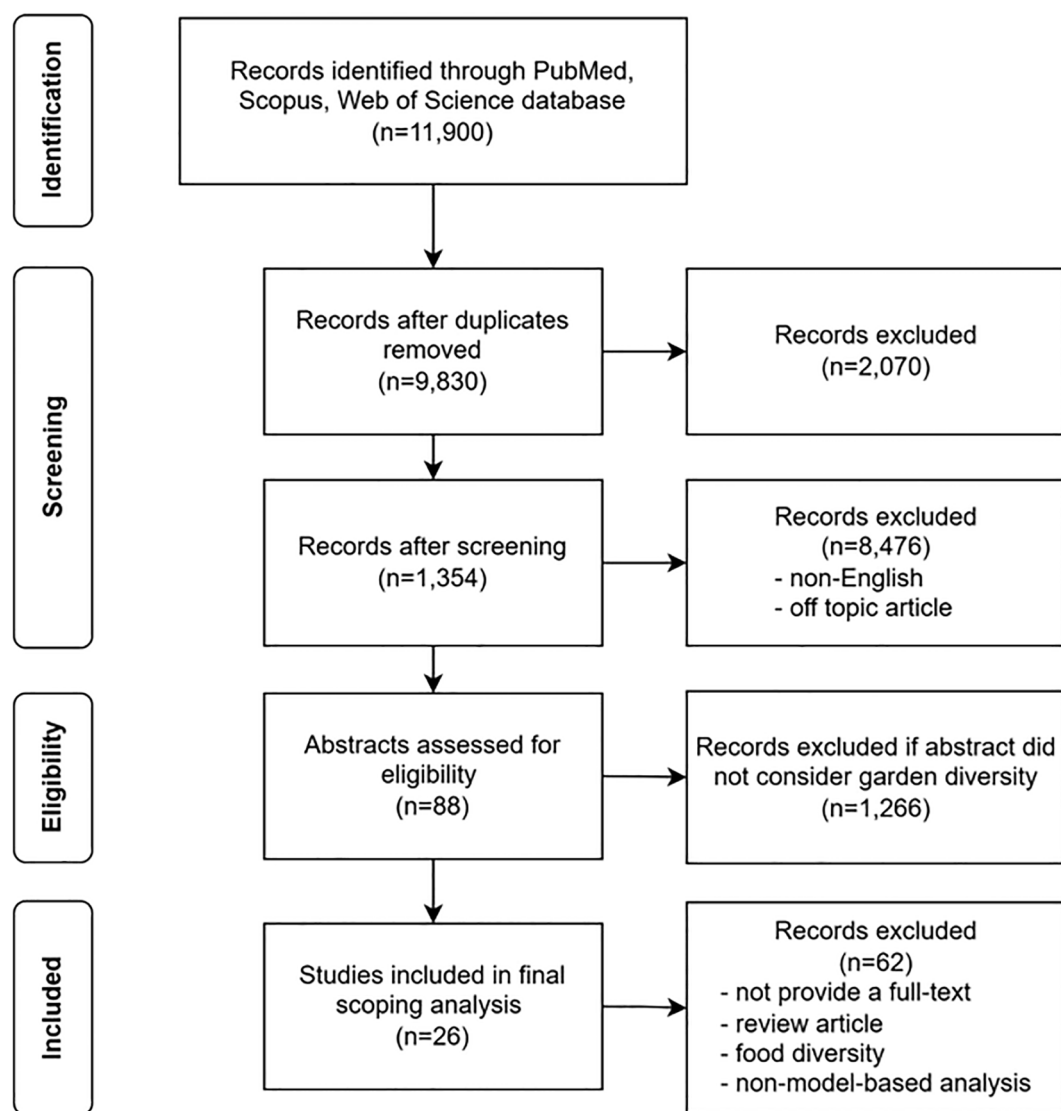


FIGURE 1

The PRISMA flow diagram summarizes the selection of 26 core studies (2000–2022). An additional four eligible studies published between 2023 and 2025 were identified in a supplementary update search and are included in the synthesis but not shown in this flow diagram.

Corporation, Redmond, WA, USA). Rather than estimating pooled effect sizes, we report the direction of statistically significant relationships (positive vs. negative) between garden attributes and biodiversity outcomes, as visualized in [Figures 2, 3](#).

3 Results

3.1 Summary of study subjects and survey items

This review synthesizes findings from 30 research papers, including four supplementary papers published between 2023 and 2025, to evaluate the ecological benefits of urban gardens and the influence of both physical environmental conditions and socioeconomic factors related to garden management on

biodiversity across various taxa. For each study, data were systematically extracted regarding research sites, biodiversity assessment methods, garden characteristics, surrounding landscape conditions, human-related variables, and statistical models used. To facilitate interpretation, abbreviations were assigned to each element and are presented in [Table 1](#). [Table 2](#) provides an overview of the variables investigated in each study using the corresponding abbreviations.

In total, 30 studies published between 2014 and 2025 met our inclusion criteria ([Figure 4](#)). The number of analyzed publications increased over time and peaked in 2021, when nine eligible articles were published, followed by five in 2022. For studies published between 2014 and 2022 we applied the full systematic search strategy described in Section 2.1. By contrast, studies from 2023–2025 were identified through a targeted supplementary search that was intended to capture obvious recent additions rather than to

TABLE 1 Summary and abbreviation of research elements.

Group	Abbreviation	Name	Details
Models used in analysis	M	Mixed-effect models	LMEM, GLMM, GLMEM, GAM, GAMM
	R	Regression models	Regression trees, linear regressions, non-linear regressions, GLM, GLM with Poisson, Binary logit model
	S	Structural equation models	Multilevel SEM, Parallel mediation models
	O	Categorical and ordinal data models	Mixed-effect ordinal regression models
Types of garden	PU	Public garden	Domestic garden, urban garden
	PR	Private garden	Home garden, residential garden, rooftop garden, facade garden, private garden
	AG	Agroforestry garden	Agroforestry garden
	AL	Allotment garden	Allotment garden
	CO	Community garden	Community garden
Measurement of diversity	SR	Species richness	Variety of different species present
	SA	Species abundance	Quantity of individuals of a species
	SD	Species density	The number of species in an area
	SC	Species classification	Native, local, exotic, rare, vulnerable, threatened
	FD	Functional diversity	Include information on functional traits among species
	MD	Morphological diversity	The variation of observable characteristics within biological groups
	PD	Phylogenetic diversity	Phylogenetic relationships among species
	DI	Diversity indices	Shannon, Shannon-wiener, Shannon-weaver, Shannon evenness, Simpson index
Landscape analysis	LO	Local factor	Percentage of ground cover (bare, grass, plant, mulch), vegetation, garden size, sun and shade hour
	LA	Landscape factor	Distance to field (forest, grassland)/street (urban)/water, concrete and building cover
	EN	Environmental factor	Soil characteristics, urban intensity gradient, structural diversity, degree of urbanization
Social analysis	E	Socioeconomic status	Job, income(wage), wealth, proportion of food consumption, land holding
	D	Sociodemographic status	Age, gender, nationality, education level, language spoken, family size, distance from garden
	P	Perceptions	Gardener experience (years of gardening, hours spent in garden), well-being, biodiversity, species preference
	MP	Management practice	Digging, cutting, composting, harvesting, fertilizer, pesticides, watering, weeding

replicate the full PRISMA-ScR procedure. Consequently, the comparatively low numbers for 2023–2025 in [Figure 4](#) should be interpreted with caution, as they likely underestimate the true volume of relevant studies published in these years. Geographically, the studies were mostly conducted in Europe (33.3%), followed by Asia (26.7%), and North America (23.3%), with fewer studies from Oceania and South America (6.7% each), and a single study from Africa (3.3%) ([Figure 5](#)). In terms of garden types, private gardens were the most frequently studied overall (40.0%), followed by public gardens (28.6%), community and agroforestry gardens (11.4% each), and allotment gardens (8.6%).

However, the distribution of garden types differed markedly among regions ([Figure 6](#)). European studies predominantly focused on private gardens, but also included several public and allotment gardens, whereas research in North America was concentrated on public and community gardens with comparatively few private gardens. Asian studies covered a more balanced mix of garden types, including public and private gardens as well as agroforestry and allotment systems. By contrast, the studies from South America, Africa and Oceania mainly examined single garden types (mostly private or agroforestry gardens), indicating regional biases in the kinds of urban gardens represented in the current

TABLE 2 The analysis information for each paper organized by taxonomic group.

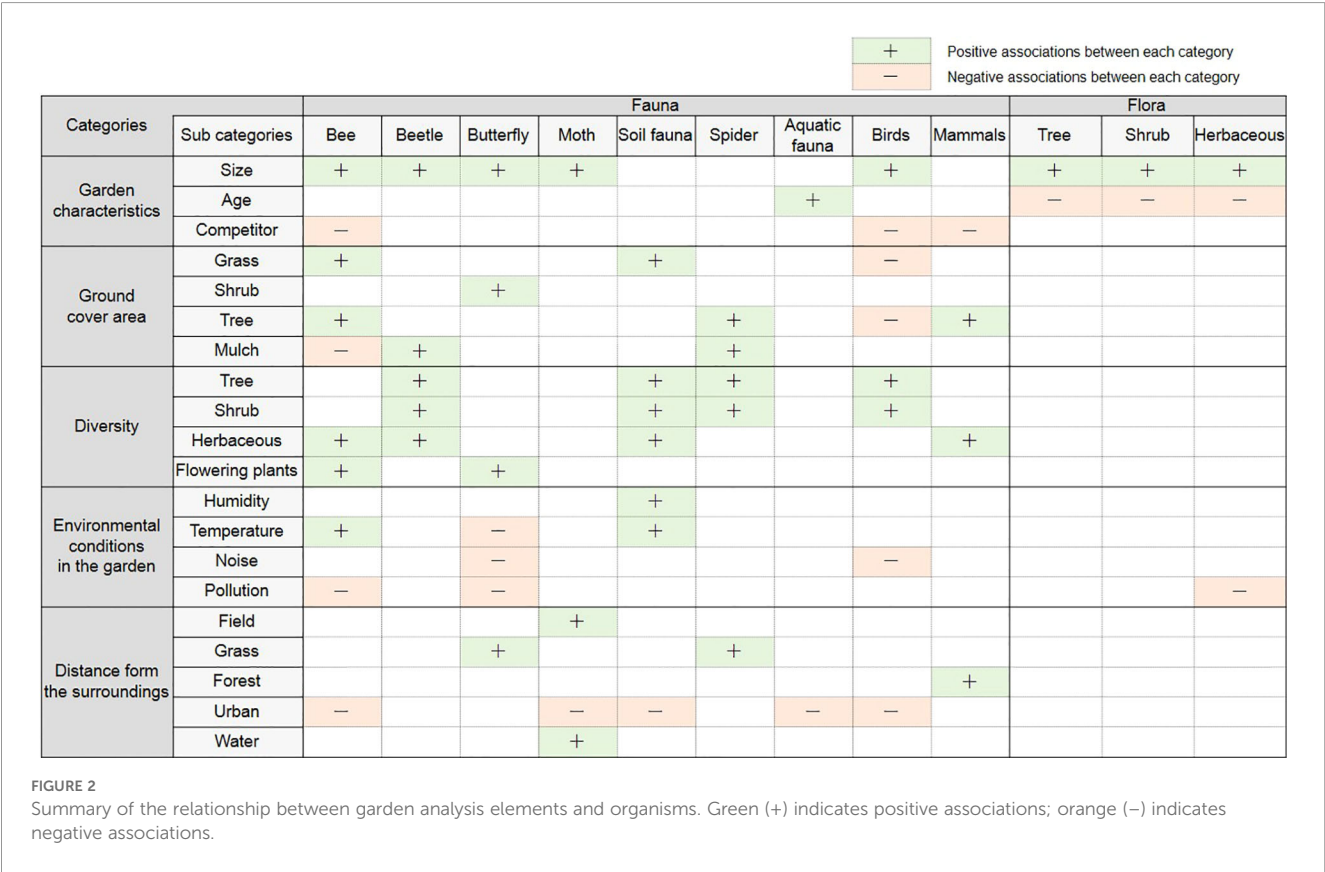
Taxonomic group	Garden type	Nations	Model	Diversity	Landscape	Social	Reference
Bee	PR	USA	M	SR, SA, SC	LO, LA		Braatz et al. (2021)
	PU	USA	R	DI	LO		Plascencia and Philpott (2017)
	PU, AL	UK	M	DI	LO		Griffiths-Lee et al. (2022)
	CO	USA	M	SR, SA	LO	D, P	Ong et al. (2022)
	CO	USA	R	SA, SR, MD	LO, LA		Quistberg et al. (2016)
	PR	Netherlands	M	SR, DI	LO		Morpurgo et al. (2024)
Beetle	PR	Chile	M	SR, SA, FD	EN	D, MP	Ibarra et al. (2021)
	PR	India	M	SR, SA, FD, DI	LO		Asha et al. (2021)
	PR	USA	M	SR, SA, SC	LO, LA		Braatz et al. (2021)
	PR	Switzerland	R	SR, SA, DI	LA		Braschler et al. (2020)
	CO	USA	M	SR, SA	LO, LA		Liere and Cowal (2024)
	PR	Netherlands	M	SR, DI	LO		Morpurgo et al. (2024)
Butterfly	PR	USA	M	SR, SA, SC	LO, LA		Braatz et al., 2021
	PU	Singapore	R	SR, DI	LO, EN	MP	Wang et al. (2017)
	PU	Hong Kong	M	SR	LO		Lo et al. (2021)
	PR, PU	UK	S	SR	EN	P	Fisher et al., 2022
Moth	PU	England	M	SR, SA, SC	EN		Bates et al. (2014)
Soil fauna	PR, AL	Switzerland	S, M	SA, FD, PD, DI	EN	MP	Tresch et al. (2019b)
	PR, AL	Switzerland	S, M	SA, DI	EN	MP	Tresch et al. (2019a)
	PR	Switzerland	R	SR, SA, DI	LA		Braschler et al. (2020)
	CO	USA	M	SR, SA	LO, LA		Liere and Cowal (2024)
	PR	Netherlands	M	SR, DI	LO		Morpurgo et al. (2024)
Spider	PR	Switzerland	R	SR, SA, DI	LA		Braschler et al. (2020)
	PU	USA	R	SD, SR, MD	LO, LA		Otoshi et al. (2015)
	CO	USA	M	SR, SA	LO, LA		Liere and Cowal (2024)
Plants	PR	USA	M	SR, SA, SC	LO, LA		Braatz et al., 2021
	PR	New Zealand	R	SR, SC	EN	E	Quan et al. (2021)
	PR	Mexico	R	SR, DI	LO	E, P	Castañeda-Navarrete (2021)
	PR	Nepal	R	DI		E, P	Rawal and Thapa (2022)
	PR, PU	UK	S	SR	EN	P	Fisher et al. (2022)
	PU	Israel	R	SC, DI	EN		Paker et al. (2014)
	CO	USA	M	SR, SA	LO	D, P	Ong et al. (2022)
	CO	USA	R	SA, FD	LO	E, D, P	Clarke and Jenerette (2015)
	AG	Ethiopia	R	SD, SC, DI		E, MP	Legesse and Negash (2021)
	AG	Middle Asia	R	SD		D, P	Vinceti et al. (2022)
	AG	Benin	R	SR, FD		E, D	Gbedomon et al. (2017)

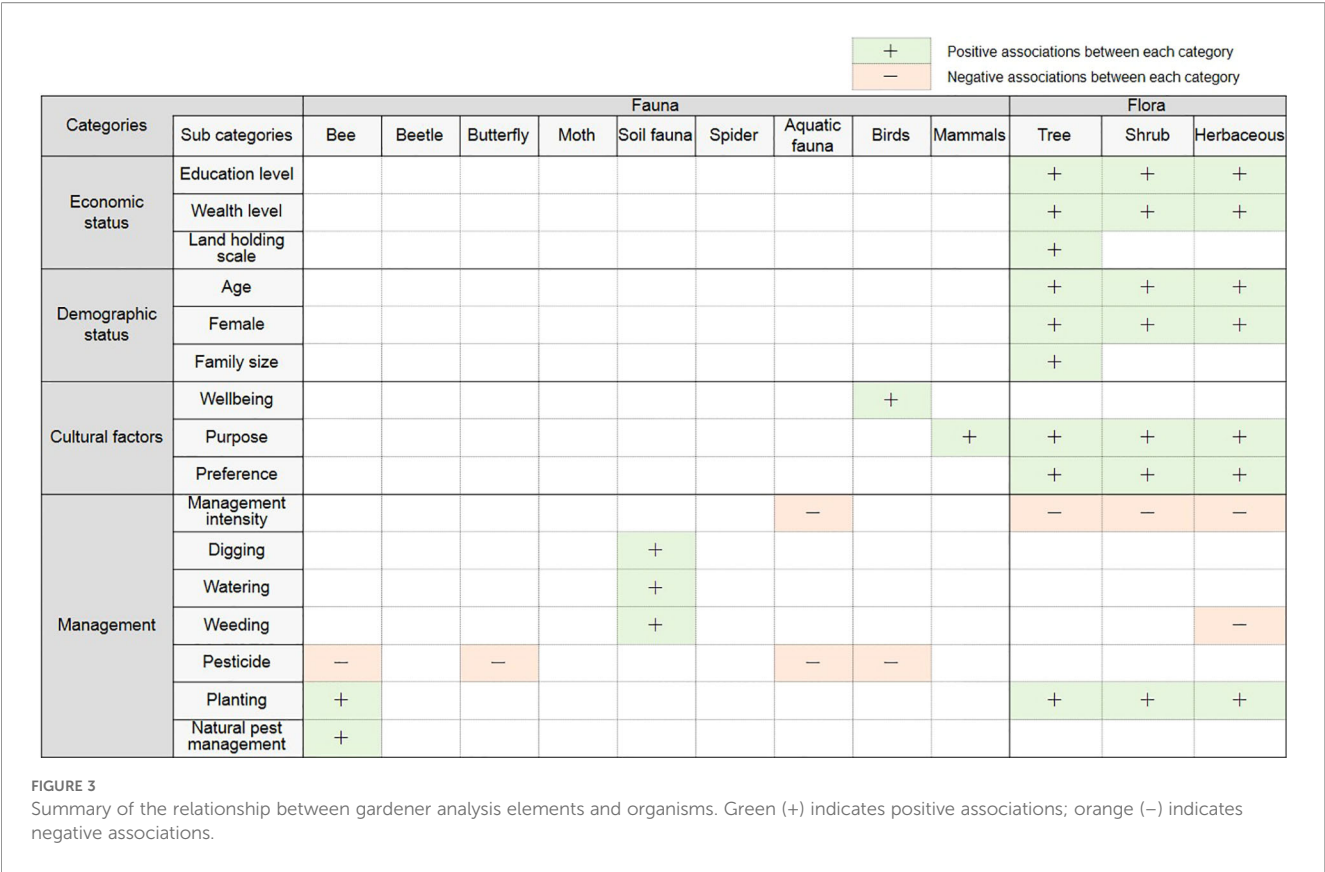
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TABLE 2 Continued

Taxonomic group	Garden type	Nations	Model	Diversity	Landscape	Social	Reference
Bird	PR	UK	O	SR, SA		D, P	Cox and Gaston (2015)
	PU	Singapore	R	SR, DI	LO, EN	MP	Wang et al. (2017)
	PU	Israel	R	SC, DI	EN		Paker et al. (2014)
	CO	USA	M	SR, SA	LO	D, P	Ong et al. (2022)
	PR, PU	UK	R	SR, SA	EN, LA	MP	de Montaigne and Goulson (2023)
	PR	Hungary	R	SR, SA, FD	EN	MP	Márton et al. (2025)
Mammal	PR	Mexico	R	SR, DI	LO	E, P	Castañeda-Navarrete (2021)
	PR	Australia	R	SA	LO		van Helden et al. (2021)
	AG	Indonesia	M	SR	LO, LA	MP	Campera et al. (2021)
Aquatic fauna	PR	Hungary	R	SR, SA, FD	EN	MP	Márton et al. (2025)

evidence base. The biodiversity assessments encompassed a wide range of taxa, including bees and beetles (13.6% each), butterflies (9.1%), moths (2.3%), soil organisms and spiders (6.8% each), birds (13.6%), mammals (6.8%), aquatic faunas (2.3%), and plants (25.0%). Among these, invertebrates were the most frequently investigated group. Furthermore, 43.3% of the studies conducted field-based assessments of garden physical environments, while 16.7% included analyses of the socioeconomic characteristics of gardeners and owners. The remaining 40.0% of the studies used a combination of physical and socioeconomic analyses. The primary statistical methods used were regression models (45.8%) and mixed-effects models (43.8%). Across the studies, biodiversity was quantified using a variety of approaches, including species richness, abundance counts, diversity indices (e.g., Shannon, Simpson), and functional trait metrics. Statistical analyses ranged from generalized linear and mixed-effects models to ordination and multivariate techniques (Table 2). This methodological heterogeneity allowed us to capture a broad range of biodiversity responses, but it also





limited the degree to which effect sizes could be directly compared or statistically synthesized across studies.

3.2 Effects of garden and environmental conditions on biota

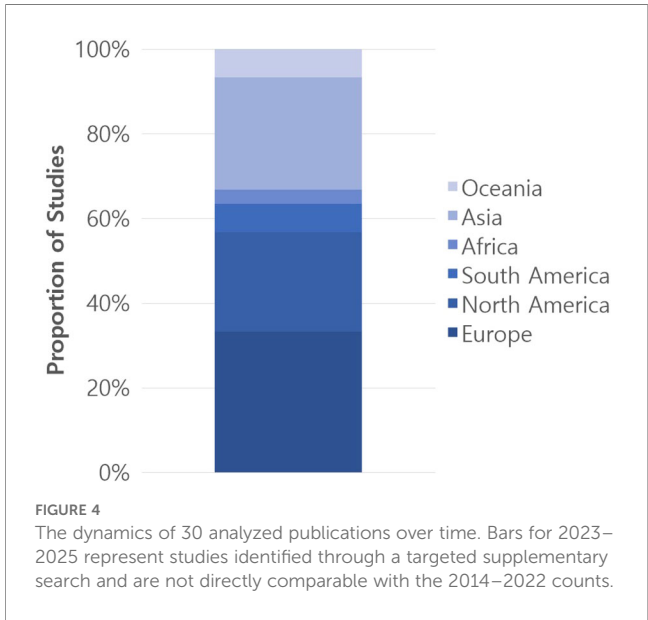
Figure 2 summarizes the statistically significant results ($p < 0.05$) from the model-based analyses examining the relationships between garden features (identified via field surveys) and biodiversity outcomes. A green “+” indicates a positive correlation, while an orange “–” represents a negative correlation. The following paragraphs are additional explanation to significant categories and associations abbreviated in the Figure 2 and then highlight additional significant relationships reported in the literature, including findings from the four supplementary studies (Supplementary Table S2).

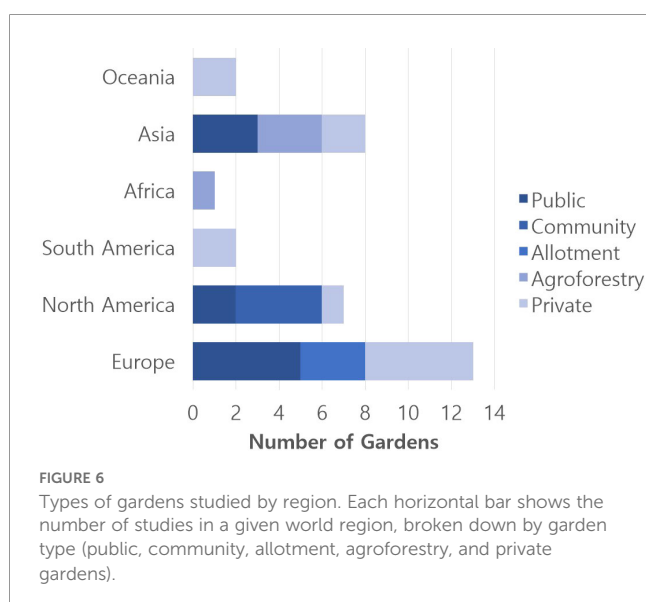
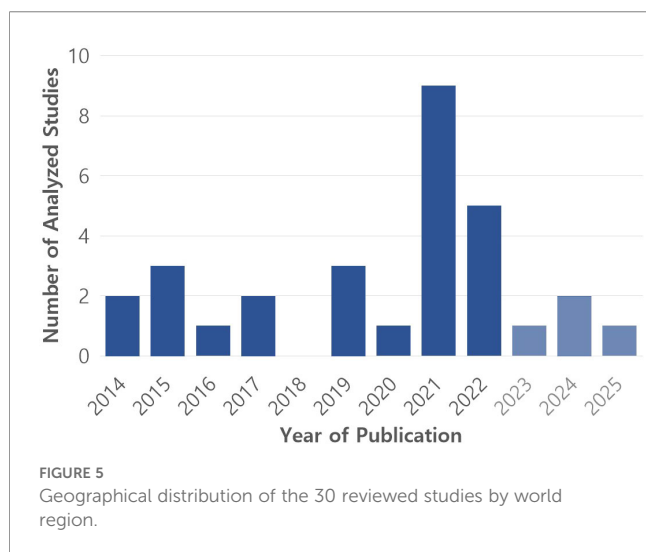
Garden size was positively associated with the diversity of bees (Quistberg et al., 2016), beetles (Ibarra et al., 2021), butterflies (Wang et al., 2017), moths (Bates et al., 2014), birds (Ong et al., 2022), and plants (Legesse and Negash, 2021). Larger gardens tend to offer a greater variety of microhabitats, which benefits a wider range of organisms (Bates et al., 2014). The diversity of rare plants was observed to decrease (Ong et al., 2022). The age and area of the pond within the garden have a significant impact on the occurrence of aquatic fauna (Márton et al., 2025).

An increase in the area of mini-meadow herbaceous plants within gardens enhanced the diversity of bees (Griffiths-Lee et al., 2022; Quistberg et al., 2016) and soil organisms (Tresch et al.,

2019b), but negatively affected bird diversity (Paker et al., 2014). Canopy coverage by shrubs and trees was positively associated with butterflies (Wang et al., 2017), bees (Ong et al., 2022), spiders (Otoshi et al., 2015), and mammals (van Helden et al., 2021), but negatively associated with birds (Paker et al., 2014).

The diversity of woody and herbaceous plants showed varying associations with different taxa, including beetles (Tresch et al., 2019a), spiders (Otoshi et al., 2015), soil fauna (Braschler et al.,





2020; Liere and Cowal, 2024), birds (Paker et al., 2014; Wang et al., 2017), bees (Quistberg et al., 2016), and mammals (van Helden et al., 2021). A high diversity of flowering plants particularly promoted the richness of bees (Braatz et al., 2021; Griffiths-Lee et al., 2022; Morpurgo et al., 2024; Quistberg et al., 2016) and butterflies (Wang et al., 2017). Rare plants (organism as rare based on frequency of occurrence) were found to have cascading positive effects on rare taxa such as bees and birds, indicating inter-taxa dependencies within gardens (Ong et al., 2022).

The temperature and humidity conditions within the garden influenced bees (Griffiths-Lee et al., 2022), butterflies (Lo et al., 2021), and soil organisms (Tresch et al., 2019a; 2019b). In contrast, environmental stressors such as noise and traffic-related pollution negatively affected butterflies, birds, and bees (Fisher et al., 2022; Wang et al., 2017).

Proximity to natural features such as fields, water bodies, and forests positively influenced the diversity of taxa including moths (Bates et al., 2014), butterflies (Lo et al., 2021), spiders (Otoshi et al.,

2015), birds (de Montaigne and Goulson, 2023), and mammals (Campera et al., 2021). Conversely, urban intensity and impervious surface coverage were negatively associated with the diversity of bees (Quistberg et al., 2016), moths (Bates et al., 2014), soil fauna (Tresch et al., 2019a; 2019b), birds (de Montaigne and Goulson, 2023), aquatic fauna (Márton et al., 2025), and plants (Ong et al., 2022).

3.3 The impact of human social conditions in gardens on biota

Figure 3 illustrates the statistically significant associations ($p < 0.05$) derived from the models analyzing the relationship between gardener's or garden owner's characteristics, based on survey data, and the organisms within the gardens. The subsequent paragraphs elaborate on these patterns (Figure 3) and incorporate further evidence from the supplementary studies (Supplementary Table S2), providing a narrative synthesis of how social conditions and management practices shape biodiversity.

Higher levels of education (Rawal and Thapa, 2022), wealth (Quan et al., 2021), income (Clarke and Jenerette, 2015), landholding size (Legesse and Negash, 2021), and household size (Legesse and Negash, 2021) were generally associated with increased plant diversity. Older and female gardeners showed a greater preference for rare and diverse plant species (Gbedomon et al., 2017; Ong et al., 2022; Rawal and Thapa, 2022), suggesting that sociodemographic background significantly shapes garden management and species composition.

Frequent garden maintenance activities such as weeding, soil disturbance, and digging were negatively associated with plant and soil organism diversity (e.g., earthworms and collembolans). In contrast, gardens with a higher proportion of perennial crops exhibited greater soil biodiversity (Tresch et al., 2019a). Pesticide use was negatively linked to the diversity and health of bees, butterflies, and birds (de Montaigne and Goulson, 2023; Griffiths-Lee et al., 2022; Wang et al., 2017), highlighting the need for alternative natural pest control strategies (Ibarra et al., 2021). Algaecides had a negative effect on the presence of aquatic fauna (Márton et al., 2025).

Cultural identity and preferences were reflected in plant species composition (Clarke and Jenerette, 2015; Legesse and Negash, 2021; Vinceti et al., 2022). Private or agroforestry gardens established for food production purposes were more likely to harbor a diverse array of plants and animals (Braatz et al., 2021; Castañeda-Navarrete, 2021; Legesse and Negash, 2021; Rawal and Thapa, 2022). Management styles varied across garden ownership types and were influenced by socioeconomic factors, thereby shaping overall plant patterns (Clarke and Jenerette, 2015). Community participation and support were especially effective in enhancing biodiversity in low-income areas (Clarke and Jenerette, 2015).

4 Discussion

In this section, we first discuss how physical environmental factors in and around gardens influence biodiversity (4.1). We then

examine how gardeners' socioeconomic characteristics and management practices shape biodiversity outcomes (4.2). Finally, we explore how environmental and human factors interact, including trade-offs and synergies for urban biodiversity management (4.3).

4.1 Physical environmental factors and species richness of gardens

The physical characteristics of gardens exert a direct influence on biodiversity. Larger gardens tend to support a wider range of microhabitats, which in turn provide a conducive environment for diverse taxa including insects, birds, and plants. The size and structural diversity of gardens function as key drivers of species richness, consistent with findings from Beninde et al. (2015). However, recent studies suggest that the importance of garden size may vary with spatial scale and urban context. For example, Morpurgo et al. (2024) found that in small private urban front gardens, garden area itself was not a significant predictor of insect diversity, whereas vegetation density and plant species richness were the primary determinants. This indicates that once a minimal habitat threshold is reached, vegetation quality and structural complexity may outweigh the effect of area. Thus, while garden size remains a major driver of biodiversity at broader landscape scales, in highly compact urban settings, optimizing vegetation density can effectively compensate for limited space.

Compared to semi-natural habitats, gardens typically maintain an intermediate level of species diversity (Thompson et al., 2003). However, species accumulation curves in urban gardens do not reach a saturation point, suggesting their potential for continued biodiversity enrichment (Loram et al., 2008). Private gardens constitute a substantial portion of urban green space, with their size and structure varying across cities (Loram et al., 2007). Despite their limited scale, these gardens collectively contribute to urban biodiversity by forming interconnected networks that support wildlife habitats (Goddard et al., 2010). In spatially constrained urban gardens, limited area can impose an upper limit on species richness, although habitat heterogeneity and appropriate management may partially overcome this limitation (Pinilla-Rosa et al., 2023).

With respect to rare taxa in urban community gardens, Ong et al. (2022) reported that more than half of all plant cultivars recorded were rare, yet fewer rare plants were found in older gardens and in gardens with more bare soil. This is possibly due to more intensive and organized maintenance by interns and volunteers. In Mexican home gardens, overall plant diversity was higher in younger gardens, particularly where owners actively exchanged plant material with other people, indicating that social practices can override simple age effects (Aguilar-Støen et al., 2009). Independent of age, plant species richness also tends to be higher in ecologically managed gardens than in conventionally managed ones (Lindemann-Matthies and Marty, 2013). Together, these findings suggest that chronological garden age alone is a poor predictor of both overall and rare plant diversity. This finding suggests that the

ongoing management of gardens and the management personnel structure may have a limited impact on the maintenance of rare plant species. It also highlights the need for further research to understand the impact of garden management on biodiversity. Individual garden components such as diverse herbaceous area, shrubs, and trees offer specialized habitats for different taxa. For instance, increased grass area benefits pollinators and soil organisms but may negatively impact bird diversity. These findings highlight the complexity of structural heterogeneity in shaping biodiversity outcomes. Since the impacts of vegetation structure vary across taxa, plant selection strategies must be tailored based on the desired conservation outcomes.

Plant composition in urban gardens is influenced by multiple variables including climate, latitude, land-use types, economic conditions (Guzo et al., 2024), elevation, traffic density, and topography (Zhang et al., 2020). Gardens with a higher proportion of native species are generally more supportive of biodiversity compared to those dominated by exotics (Davies et al., 2009; Salisbury et al., 2017).

Moreover, gardens function as critical pollinator habitats that can enhance fruit and seed production in both cultivated and wild flora (Cussans et al., 2010). Concurrently, urban gardens function as environments that can support a diverse array of biota. Urban community gardens contain diverse wild bee populations due to their high functional attribute diversity (Normandin et al., 2017). Plant species richness, vegetation complexity, and the availability of nesting resources such as deadwood and bare ground are critical for supporting diverse urban bee communities (Ayers and Rehan, 2023; Felderhoff et al., 2023). These factors enhance both taxonomic and functional diversity, compensating for spatial constraints in cities, while impervious surfaces negatively affect bee populations (Kaiser and Resasco, 2024). Additionally, diverse plantings can support other beneficial insects (Nestle et al., 2020). This finding indicates that urban gardens serve a function beyond mere green spaces and can act as vital habitats for a diverse array of species.

However, floral abundance may have guild-specific effects; for instance, it increased ground-dwelling spider activity (Otoshi et al., 2015) but reduced species richness for foliage-dwelling spiders (Liere and Cowal, 2024), suggesting that vegetation structure influences predators in nonlinear, taxon-dependent ways. Moreover, high plant species richness does not always translate into positive biodiversity outcomes. In highly productive urban habitats, resource enrichment and reduced predation can favor a few dominant or invasive species, leading to competitive exclusion and reduced community evenness (Shochat et al., 2010). Such processes illustrate how increased productivity and homogenized resource inputs may paradoxically drive biodiversity loss through intensified interspecific competition. This suggests that there may be thresholds of optimal diversity, beyond which ecological benefits plateau or even decline, reinforcing the importance of functional rather than purely taxonomic diversity in garden design.

Although urban ponds generally harbor lower species richness than natural lakes, they play a crucial role in maintaining functional diversity within urban landscapes. Effective management—such as enhancing habitat heterogeneity and controlling invasive species—

can substantially improve their ecological value (Márton et al., 2025; Pinilla-Rosa et al., 2023).

The spatial arrangement of urban gardens also has implications for maintaining ecological continuity. The role of urban community gardens in green networks is important, and if they are lost to urbanization, the loss of ecological continuity is expected to be significant (Di Pietro et al., 2018). The ecological value of gardens is further influenced by their spatial relationship with surrounding landscapes. Liere and Cowal (2024) further highlighted that local and landscape factors differentially influence predatory arthropods in urban agroecosystems. While ground-dwelling predators were more responsive to within-garden vegetation complexity, canopy-dwelling species were influenced by the proportion of semi-natural habitats and impervious surfaces in the surrounding landscape. This finding underscores the importance of considering both within-garden habitat structure and broader landscape connectivity when assessing urban biodiversity. Ecological continuity, therefore, is not solely a function of proximity to natural areas but also of how garden-level habitat quality interacts with the surrounding matrix.

Proximity to natural elements such as fields, forests, or water bodies positively affects the diversity of certain taxa, while urbanization and anthropogenic pollutants exert negative effects. Gardens interact dynamically with adjacent habitats, both positively and negatively, depending on the degree of ecological connectivity. However, proximity to natural areas can also facilitate the spread of invasive species, potentially disrupting native ecosystems (Marco et al., 2008; Sukopp, 2004). Horticultural species introduced into urban green spaces can escape cultivation and establish in semi-natural habitats, threatening native biodiversity (McDougall et al., 2005; Vitousek et al., 1997). Although some non-native plants may benefit pollinators (Baker et al., 2020; Staab et al., 2020), their introduction must be carefully managed to minimize ecological risks (Keller et al., 2011). To understand the full extent of these interactions, future research should adopt multi-taxa approaches to comprehensively assess the ecological functions of gardens (Braschler et al., 2021). These insights reinforce the importance of biodiversity-sensitive urban planning and management strategies.

Because most of the reviewed studies were conducted in temperate regions, the strength and direction of environmental and social drivers may vary under different climatic regimes. For example, water availability, seasonality, and extreme heat events in arid or tropical cities are likely to modify the relative importance of garden structure, plant composition, and management practices for biodiversity outcomes. These regional and climatic contingencies should be explicitly considered when transferring findings across cities and continents.

4.2 Human factors: socioeconomic conditions, management practices, and cultural influences

Urban garden biodiversity is strongly associated with the cultural backgrounds of gardeners. Gardeners often select species

that reflect their cultural identities and preferences, resulting in distinct assemblages of culturally important food and ornamental plants across ethnic and social groups (Clarke and Jenerette, 2015; Vinceti et al., 2022). In many home-garden and agroforestry systems, these choices include the maintenance of native and traditional local varieties that support household livelihoods, so that gardens function as important reservoirs of on-farm agrobiodiversity (Larios et al., 2013; Legesse and Negash, 2021; Vinceti et al., 2022). Pearsall et al. (2017) observed that while cultural diversity among gardeners does not directly enhance plant species richness, it significantly shapes plant selection and the way they maintain their gardens. Das et al. (2024) reported that the cultural ecosystem services (CES) experienced in community gardens influence gardening practices, and that the CES experienced vary depending on the structure of the garden and the gardener. Furthermore, CES experiences were found to be associated with the use of local ecological knowledge and sustainable management practices, such as integrated pest management and planting diverse species.

Socioeconomic status is a critical determinant of plant composition in urban areas (Dow, 2000; Hope et al., 2003). Previous research, as synthesized by Butterfield (2020) and Birky and Strom (2013), suggests that community gardens in low-income and ethnic minority communities have often been developed to resist disinvestment and improve access to healthy food, whereas more affluent and highly educated, predominantly white communities have more recently embraced community gardening as a way to transform local food systems and address sustainability concerns. Recent evidence from Central and Eastern Europe, however, suggests that these class- and region-based distinctions in gardening motivations are beginning to blur, as food self-provisioning is gaining importance in Western European countries and biodiversity-friendly, recreational gardening is increasingly adopted in Central and Eastern Europe (Varga-Szilay et al., 2025). High education, income, and wealth levels are associated with the selection and maintenance of diverse plant species, which positively influences the diversity of associated taxa such as pollinators and birds. These discrepancies underscore the presence of socioeconomic disparities in the realm of urban greening initiatives, thereby underscoring the necessity for policy interventions that are designed to facilitate equitable access to plant resources. The enhancement of awareness regarding habitat improvement (Braschler et al., 2020) and the propensity of individuals to cultivate or maintain a diverse array of plant species on their land holdings for various purposes and service needs are projected to foster increased plant diversity in gardens (Griffiths-Lee et al., 2022; Legesse and Negash, 2021; Vinceti et al., 2022).

Conversely, excessive management practices such as frequent weeding or soil disturbance may negatively impact certain taxa. For instance, de Montaigne and Goulson (2023) demonstrated that bird abundance and species richness in UK gardens were strongly influenced by management intensity. While structurally diverse and high-quality habitats promoted avian diversity, the use of pesticides—particularly glyphosate and metaldehyde—

significantly diminished these benefits. This finding underscores that inappropriate chemical management can override the positive effects of vegetation structure and habitat diversity, highlighting the need for biodiversity-sensitive management approaches.

Human intervention plays a crucial role in shaping biodiversity outcomes in urban ecosystems. Deliberate, biodiversity-conscious garden management can contribute to the protection of plants and animal species (Lindemann-Matthies and Marty, 2013). As such, garden planning must go beyond mere green coverage to incorporate spatial arrangements and species interactions that enhance ecological functions. For instance, reduced spacing between plants and species mixing can enhance pest resistance (Nighswander et al., 2021). Continuous provisioning of non-crop floral resources can improve pollination services (Lowenstein et al., 2015), and institutional support such as compost or mulch provision can promote equitable access to ecosystem services (Egerer et al., 2017). Ecological gardening practices such as wildflower meadows, ponds, nesting sites for birds and bees, and decomposing wood piles can simultaneously increase species richness and enhance the perceived aesthetic quality of private gardens, indicating that ecological and aesthetic goals are not necessarily in conflict (Lindemann-Matthies and Marty, 2013).

Community engagement plays a critical role in sustaining the ecological value of gardens. Participation and support from local communities are essential not only for fostering awareness of ecological functions but also for enhancing biodiversity in marginalized neighborhoods (Davies et al., 2009). The role of the community, in conjunction with the manner in which gardens are managed, constitutes a pivotal social foundation for sustaining ecosystem health. Consequently, the implementation of programs and educational initiatives targeting local communities is imperative (van Heezik et al., 2012). This collective sense of community is anticipated to exert a favorable influence on the preservation of biodiversity. Gardens should be recognized as vital spaces for promoting food security, biodiversity conservation, and urban sustainability through public policy (da Cunha et al., 2020).

4.3 Complementary and correlations between environmental and human factors

The physical environment and social dimensions of gardens function in a complementary, rather than independent, manner. When physical factors (e.g., area, vegetation structure, landscape connectivity) are aligned with socioeconomic conditions (e.g., education, income, management attitudes), biodiversity outcomes are maximized. For example, structurally complex and large gardens offer basic habitat functions, but when coupled with owners possessing high socioeconomic capital and nature-friendly practices, they can support the diversity of more sensitive taxa such as rare species or soil fauna (Ibarra et al., 2021).

In addition to ecological benefits, garden design and management can contribute significantly to human well-being. Studies have shown that horticultural practices in urban environments positively influence both physical and mental

health (Dunnett and Qasim, 2000; Niemelä, 1999). Community gardens, in particular, enhance human-nature relationships and promote ecological awareness (Lin et al., 2018). Therefore, the conservation potential of urban gardens should not be limited to green space expansion but must be grounded in ecologically functional design and management practices.

While enhancing biodiversity is often a key objective in urban garden planning, it is not always the unequivocal ideal. Some studies point out that higher biodiversity can increase management complexity, maintenance costs, and the risk of ecological disservices, which may act as barriers to sustained gardener participation (Bassett et al., 2022; Martens et al., 2022; Roman et al., 2021). In Berlin, for instance, residents expressed a strong preference for biodiverse urban green spaces but revealed varied willingness to engage in biodiversity-supporting actions depending on their available time and socio-cultural background (Martens et al., 2022).

In urban greenspaces, the relationship between maintenance intensity and biodiversity outcomes varies substantially—different management regimes can yield contrasting results depending on context and capacity (Hu and Lima, 2024). In urban forestry, for example, attempts to maximize wildlife habitat through tree retention can conflict with safety, aesthetics, and public expectations—showing the need for context-dependent management standards (Bassett et al., 2022). These findings emphasize that biodiversity-based interventions must be aligned with realistic maintenance capacities and social acceptance.

A growing body of literature recognizes that trade-offs between biodiversity and other urban ecosystem functions or social goals are inevitable. Stijnen et al. (2024) identified governance, functional, and inclusivity trade-offs in urban nature-based solutions, while Aronson et al. (2017) emphasized that effective biodiversity management requires compromise between human perceptions, needs, and ecological requirements across multiple spatial and social scales. Similarly, Prioreschi et al. (2024) demonstrated that intensive recreation and frequent human use in urban parks can negatively affect habitat quality and species richness, highlighting a recreation–biodiversity trade-off in naturalized urban spaces. Together, these studies show that maximizing biodiversity alone can conflict with goals of social inclusion, accessibility, and aesthetic preference—key elements in sustaining public engagement.

Williams et al. (2009) likewise described how human preferences act as urbanization filters, creating non-random species assemblages and reshaping community structure depending on cultural and aesthetic norms. These selective processes underscore that urban biodiversity is co-produced by ecological dynamics and human decisions.

Social-ecological interactions also determine how biodiversity initiatives are perceived and maintained. Martens et al. (2022) showed that residents' willingness to support biodiversity-friendly actions in green spaces depends strongly on socioeconomic and cultural contexts, linking participatory motivation with resource availability. This resonates with Aronson et al.'s (2017) findings that management coordination across private gardens and public areas is often fragmented and that multi-stakeholder collaboration is essential

for coherent biodiversity outcomes. Such coordination challenges further complicate the pursuit of purely ecological objectives.

In practice, wildlife-oriented management can also generate unintended conflicts if practicality and safety are overlooked. Bassett et al. (2022) noted that pruning restrictions designed to protect nesting birds may hinder urban forestry operations, reinforcing the importance of adaptive and multi-stakeholder standards.

Consequently, biodiversity-centric management strategies should be tempered with awareness of practical, social, and ecological constraints. Urban gardens serve multiple overlapping roles—ecological, cultural, recreational, educational, and therapeutic—and maximizing species richness alone may not align with these varied functions. A more nuanced and adaptive approach, integrating biodiversity objectives with maintenance feasibility, gardener capacity, and community engagement, is more likely to yield resilient socio-ecological outcomes.

Future work should focus on identifying optimal thresholds of diversity—not “as much as possible,” but “as much as sustainable”—and on developing participatory governance models that balance ecological goals with usability, safety, and inclusivity. By acknowledging the trade-offs between biodiversity enhancement and human management realities, urban garden planning can evolve toward truly resilient, inclusive, and adaptive green systems.

5 Conclusion

This scoping review provides a comprehensive understanding of ecological value of urban gardens, highlighting their role not only as green spaces but also as important habitats that support diverse biota and enhance biodiversity in urban environments.

Ultimately, this review highlights the interdependent relationship between socio-cultural and environmental factors in shaping urban garden biodiversity. The ways in which gardens are created, managed, and utilized vary according to the socioeconomic status and cultural background of gardeners, which in turn influence ecological outcomes. These findings underscore the need for an integrative framework that views biodiversity as both a social and ecological phenomenon. Strategic planning of urban gardens should therefore incorporate both physical characteristics—such as size and spatial structure—with the sociodemographic diversity of local communities. Urban garden policies that align ecological goals with the diversity of local residents will be key to enhancing biodiversity and ecosystem resilience.

This review has several limitations. First, the number of studies specifically addressing biodiversity in urban gardens remains limited. Second, due to the diversity of methods used across studies—including different analytical models and taxa—it was not possible to conduct a fully standardized comparison. Finally, prioritizing open-access, garden-centric studies may exclude potentially relevant work—a constraint we acknowledge. Additionally, our database search relied on garden-related keywords and did not explicitly include all possible biodiversity-

related terms or garden-type descriptors, which may have led to the omission of some relevant studies. To further improve the systematic coverage and reproducibility of the evidence base, future updates could adopt an expanded set of search terms, as suggested by the reviewers.

Future research should expand the geographical and taxonomic scope and focus on long-term monitoring to identify causal relationships between garden attributes and biodiversity outcomes. In addition, future research recommendations could include the use of more comparable and consistent methodological frameworks, or at least a clearer framing that facilitates cross-study comparison. Interdisciplinary approaches will be critical to understanding the socio-ecological complexity of urban ecosystems and advancing biodiversity conservation in cities. Future research should also examine how climatic and regional contexts modulate the relative importance of environmental and social drivers, particularly in under-represented regions such as the Global South.

The following three points summarize the key findings of this scoping review:

1. Garden and environmental conditions:

The size, age, and structural components of gardens—such as meadows, shrubs, and trees—and their connectivity to surrounding natural habitats significantly influence the richness and abundance of taxa, including bees, butterflies, beetles, and birds. Urbanization and anthropogenic stressors have a negative impact on these functions, underscoring the importance of considering ecological buffer roles in garden planning and urban development.

2. Human social conditions in gardens:

The socioeconomic attributes, cultural preferences, and management styles of garden owners and caretakers are key determinants of plant composition and biodiversity within gardens. Socioeconomic capital, such as high levels of education, income, and wealth, support the inclusion of diverse and rare plant species, while community engagement can promote positive ecological outcomes even in lower-income areas. Inclusive and accessible garden design policies are essential for maximizing urban biodiversity.

3. Integrated management and policy recommendations:

Since environmental and social factors operate synergistically, maintaining and enhancing biodiversity in urban gardens must go beyond green coverage expansion. Effective garden design and management should emphasize structural complexity, nature-friendly maintenance practices, enhanced connectivity to natural habitats, and increased community engagement. This integrative approach is vital for transitioning toward sustainable urban biodiversity conservation.

Data availability statement

The original contributions presented in the study are included in the article/[Supplementary Material](#). Further inquiries can be directed to the corresponding author/s.

Author contributions

JP: Conceptualization, Data curation, Formal analysis, Methodology, Visualization, Writing – original draft. MJ: Methodology, Supervision, Writing – review & editing. HJ: Conceptualization, Writing – review & editing.

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Conflict of interest

The author(s) declared that this work was conducted in the absence of any commercial or financial relationships that could be construed as a potential conflict of interest.

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Supplementary material

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